



MATHEMATICS (EXTENSION 1)

2014 Preliminary Course Assessment Task 1

SUPPLEMENTARY EXAMINATION

Tuesday 4 March, 2014

General instructions

- Working time – 45 minutes.
- Write your answers on the answer sheet supplied, numbered page 9.
- Write using *pencil*.
- Board approved calculators may be used.
- Attempt **all** questions (Questions 1 to 30).
- At the conclusion of the examination, this question paper and the answer sheet will be collected as two bundles.
- **Paper used for scratch space is to be submitted with the question paper and not the answer sheet.**

This question paper will not be read by markers.

Section I

Mark your answers on the answer sheet provided.

Questions	Marks
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- | | |
|---|---|
| 1. What is the value of $\sqrt{\pi^2 + 5}$, correct to 2 decimal places? | 1 |
|---|---|

(A) 3.85	(B) 3.86	(C) 8.14	(D) 8.15
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- | | |
|--|---|
| 2. Which of the following is 339 400 in scientific notation, correct to two significant figures? | 1 |
|--|---|

(A) 3.3×10^5	(C) 3.4×10^5
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(B) 3.3×10^6	(D) 3.4×10^6
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- | | |
|--|---|
| 3. Which of the following is the fully factored form of $4a^2y - 10ay^2$? | 1 |
|--|---|

(A) $2ay(2y - 5a)$	(C) $2a^2y^2(2y - 5a)$
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(B) $2ay(2a - 5y)$	(D) $4ay(y - 2.5a)$
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- | | |
|---|---|
| 4. Eric and Chris paid \$315 for a meal at a restaurant, which included a 12.5% credit card surcharge on top of the cost of the meal. | 1 |
|---|---|

Which of the following gives the price of the meal without the credit card surcharge?

(A) $315 - \frac{87.5}{100}$	(C) $\frac{315 \times 112.5}{100}$
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(B) $315 - \frac{112.5}{100}$	(D) $\frac{315 \times 100}{112.5}$
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- | | |
|---|---|
| 5. A student writes the following solution to express $0.\overline{271}$ as the simplest fraction. Which line does the error occur? | 1 |
|---|---|

$$\begin{array}{ll} \text{Let } x = 0.2717171 \dots & (1) \\ 10x = 2.7171717 \dots & (2) \\ 1000x = 271.717171 \dots & (3) \\ x = \frac{269}{999} & (4) \end{array}$$

(A) Line 1	(B) Line 2	(C) Line 3	(D) Line 4
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6. What does $L + 2m - (L - 2n) - [2m + L - (2n - L)]$ simplify to? 1
- (A) $4n + 2L$ (C) $4m + 4n - 2L$
(B) $4n - 2L$ (D) $4m - 4n - 2L$
7. Which of the following is equivalent to $2x^2 + 5x - 3$? 1
- (A) $(2x - 1)(x + 3)$ (C) $(2x + 3)(x - 1)$
(B) $(2x + 1)(x - 3)$ (D) $(2x - 3)(x + 1)$
8. What does $\frac{x^2 + 4x}{x^3 - 9x} \div \frac{x^2 + 2x - 8}{x^2 + x - 6}$ simplify to? 1
- (A) 1 (B) $\frac{x}{x - 3}$ (C) $\frac{1}{x - 3}$ (D) $\frac{1}{x + 3}$
9. Find the ratio of $x : y$ if $6y + 2x = -3x + 13y$. 1
- (A) 5 : 7 (B) 6 : 2 (C) 13 : -3 (D) 7 : 5
10. What is the exact form solution to $2x^2 - 5x - 2 = 0$? 1
- (A) $x = \frac{-5 \pm 3}{4}$ (C) $x = \frac{-5 \pm \sqrt{41}}{4}$
(B) $x = \frac{5 \pm 3}{4}$ (D) $x = \frac{5 \pm \sqrt{41}}{4}$
11. What are the values of a and b if $\frac{5 - 2\sqrt{2}}{1 + \sqrt{2}} \equiv a + b\sqrt{2}$? 1
- (A) $a = -9, b = 7$ (C) $a = -7, b = 9$
(B) $a = 9, b = -7$ (D) $a = 7, b = -9$
12. What is $8^3 \times 6^{\frac{1}{2}} \div 32^{\frac{3}{2}}$ in simplest form? 1
- (A) $4\sqrt{3}$ (B) $2\sqrt{3}$ (C) $3\sqrt{2}$ (D) $4\sqrt{2}$
13. $\frac{1}{\sqrt[4]{m^3}}$ in index form is 1
- (A) $\frac{3}{m^{\frac{1}{4}}}$ (B) $m^{-\frac{3}{4}}$ (C) $m^{\frac{3}{4}}$ (D) $m^{\frac{4}{3}}$

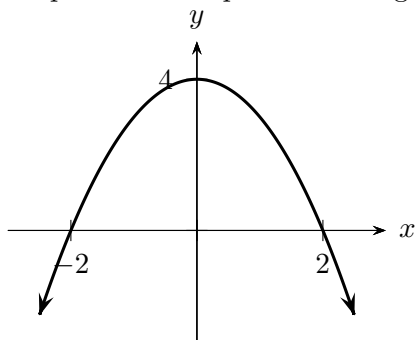
14. If $4^{x-1} = 32$, what is the value of x ? 1

(A) 10 (C) 3
(B) 3.5 (D) 6

15. Which of the following represents the fully factored form of $64p^3 + 8$? 1

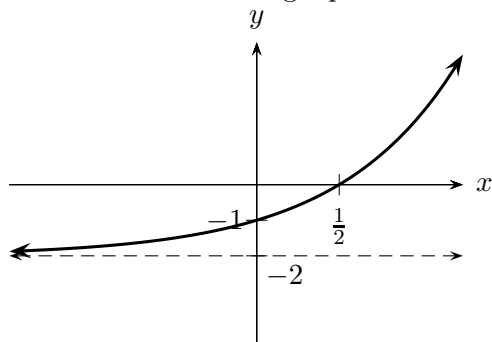
(A) $(4p + 2)(16p^2 - 8p + 2)$ (C) $8(2p + 1)(4p^2 - 2p + 1)$
(B) $(4p + 2)(16p^2 + 8p + 2)$ (D) $8(2p + 1)(4p^2 + 2p + 1)$

16. Which equation corresponds to the graph below? 1



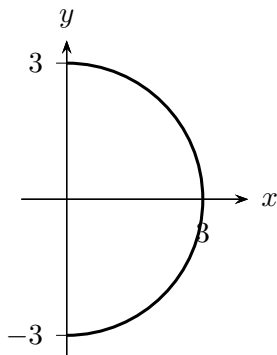
(A) $y = 2 - x^2$
(B) $y = 4 - x^2$
(C) $y = x + 4$
(D) $y = x^2 + 4$

17. Which of the following equations best describes the graph shown? 1



(A) $y = 4^x - 1$
(B) $y = 2^x - 1$
(C) $y = 4^x - 2$
(D) $y = 2^x - 2$

18. Which of the following represent the equation of the semicircle shown below? 1



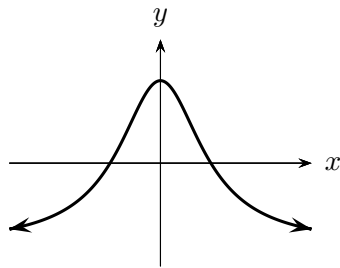
(A) $y = \sqrt{9 - x^2}$
(B) $y = -\sqrt{9 - x^2}$
(C) $x = \sqrt{9 - y^2}$
(D) $x = -\sqrt{9 - y^2}$

19. A particular relation $R(x)$ obeys the property $R(x) \neq R(-x)$, as well as $R(-x) \neq -R(x)$.

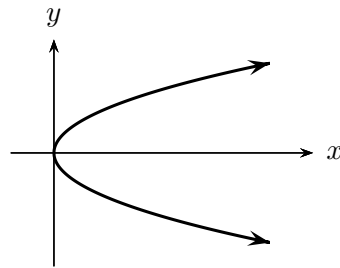
1

Which graph best represents this relation?

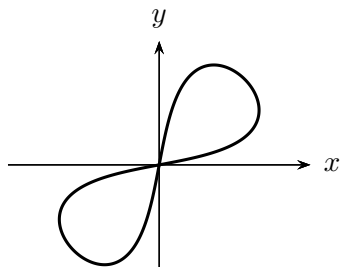
(A)



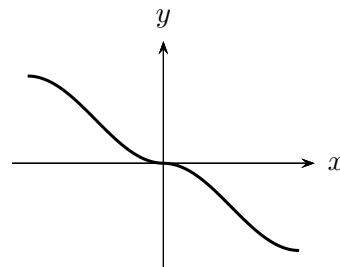
(C)



(B)



(D)

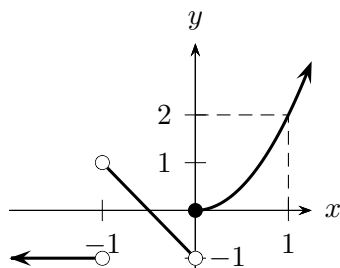


20. Which graph best describes the following piecewise-defined function

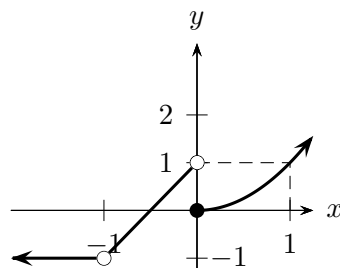
1

$$f(x) = \begin{cases} -1 & x < -1 \\ 2x + 1 & -1 < x < 0 \\ 2x^2 & x \geq 0 \end{cases}$$

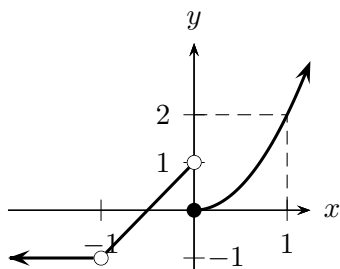
(A)



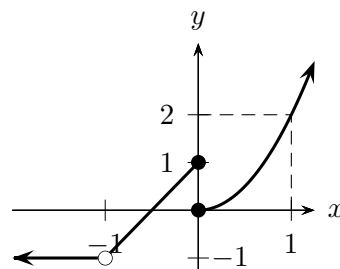
(C)



(B)

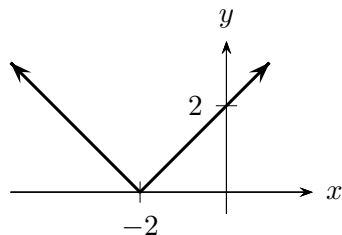


(D)

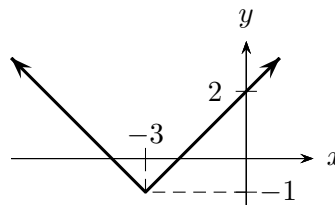


21. Which of the following graphs best represents $y = |x + 3| - 1$? 1

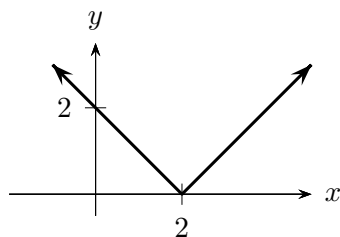
(A)



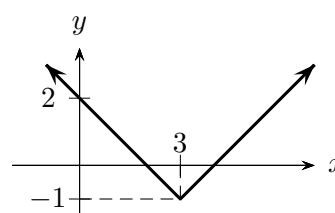
(C)



(B)



(D)



22. Which of the following best describes the function $f(x) = \frac{x^3 - 1}{x^2}$? 1

(A) Even function

(C) Neither odd nor even function

(B) Odd function

(D) Zero function

23. The graph with equation $y = x^3$ is stretched vertically by a factor of $\frac{1}{4}$, translated 4 units to the left and 3 units up. Which equation represents the resulting graph? 1

(A) $y = \frac{1}{4}(x - 4)^3 + 3$

(C) $y = \frac{1}{4}(x + 4)^3 + 3$

(B) $y = \frac{1}{4}(x - 4)^3 - 3$

(D) $y = \frac{1}{4}(x + 4)^3 - 3$

24. What is the domain of $f(x) = 3 - \sqrt{2x + 1}$? 1

(A) $x \neq -\frac{1}{2}$

(B) $x \leq -\frac{1}{2}$

(C) $x \leq \frac{1}{2}$

(D) $x \geq -\frac{1}{2}$

25. What is the value of k if $4x^2 - 6x + k$ is a perfect square? 1

(A) $\frac{4}{9}$

(B) $\frac{9}{4}$

(C) 4

(D) 9

26. Which of the following represent the solutions of $3x - x^2 < 0$? 1

(A) $0 < x < 3$

(C) $x < 0, x > 3$

(B) $x < 0, x < 3$

(D) $x > 0, x < 3$

27. Which of the following are the asymptotes to $y = \frac{3x - 2}{x + 7}$? 1

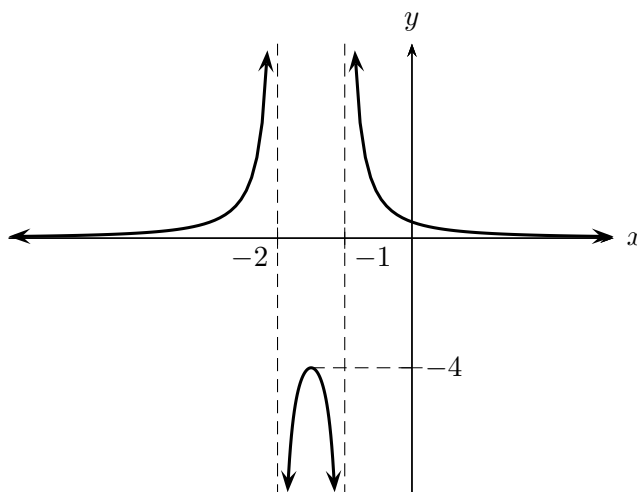
(A) $x = 3, y = 7$

(C) $y = 3, x = 7$

(B) $x = 3, y = -7$

(D) $y = 3, x = -7$

28. The graph of $y = f(x)$ is shown in the diagram below. 1



Which of the following represents the range of the function?

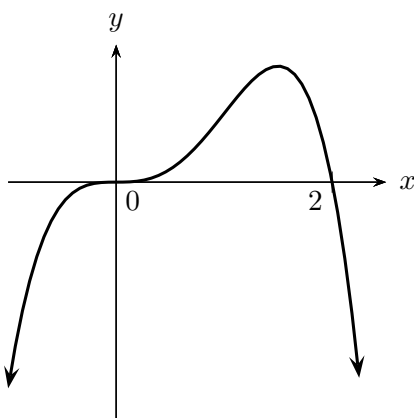
(A) $R = \{y : y > 0, y \leq -4\}$

(C) $R = \{y : y \neq -1, y \neq -2\}$

(B) $R = \{y : y \geq 0, y \leq -4\}$

(D) $R = \{y : y \in \mathbb{R}\}$

29. Which of the following could represent $y = f(x)$? 1



(A) $f(x) = x^3(2 - x)$

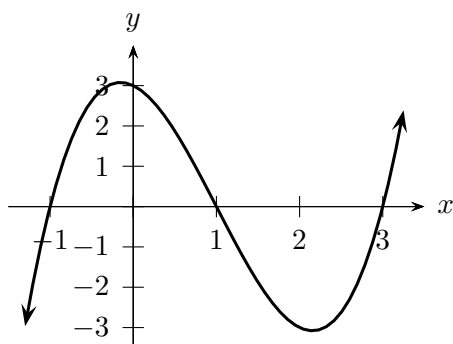
(C) $f(x) = x^3(x - 2)$

(B) $f(x) = x^2(2 - x)^2$

(D) $f(x) = -x^3(x + 2)$

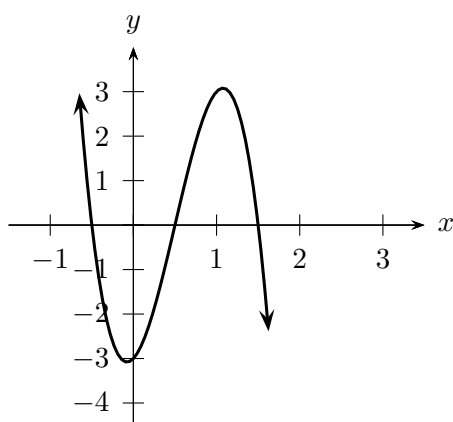
30. The graph of $y = f(x)$ is shown. The graph which best represents $1 - f(2x)$ is

1

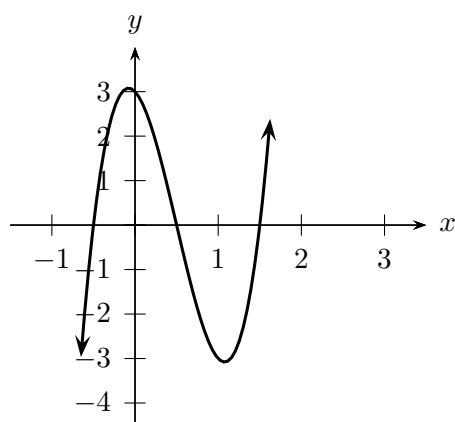


(A)

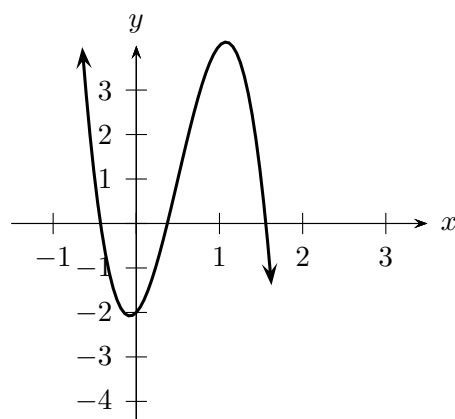
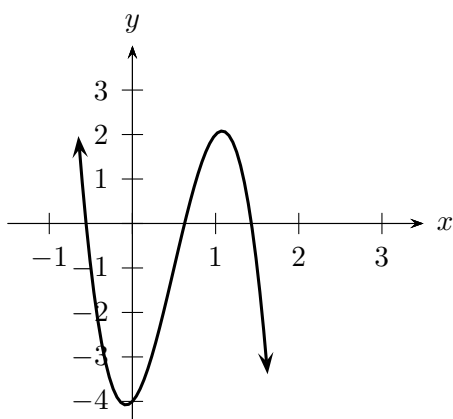
(C)



(B)



(D)



End of paper

Answer sheet for Mathematics Extension 1 Supplementary Assessment Task 1

Mark answers by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

4

Class (please ✓)

☐ 11M3A – Ms Ziaziaris

☐ 11M3B – Mr Ireland

☐ 11M3C – Mr Lin

☐ 11M3D – Mr Weiss

☐ 11M3E – Mr Berry

☐ 11M3F – Mr Lam

1 –

☐ A

☐ B

☐ C

☐ D

2 –

☐ A

☐ B

☐ C

☐ D

3 –

☐ A

☐ B

☐ C

☐ D

4 –

☐ A

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5 –

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30 –

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☐ D

Marker’s use only.

QUESTION	1-15	16-25	27-30	Total	%
MARKS	15	10	5	30	

NORTH SYDNEY BOYS' HIGH SCHOOL

TUESDAY 4 MARCH, 2014

Answer sheet for Mathematics Extension 1 Assessment Task

Mark answers by fully blackening the correct circle, e.g. "●"

SOLUTIONS

STUDENT NUMBER:

4

Class (please ✓)

☐ 11M3A – Ms Ziaziaris☐ 11M3B – Mr Ireland☐ 11M3C – Mr Lin☐ 11M3D – Mr Weiss☐ 11M3E – Mr Berry☐ 11M3F – Mr Lam

- 1 – (A) ● (C) (D)
 2 – (A) (B) ● (D)
 3 – (A) ● (C) (D)
 4 – (A) (B) (C) ●
 5 – (A) (C) ●
 6 – (A) ● (C) (D)
 7 – ● (B) (C) (D)
 8 – (A) (B) ● (D)
 9 – ● (B) (C) (D)
 10 – (A) (B) (C) ●
 11 – ● (B) (C) (D)
 12 – ● (B) (C) (D)
 13 – (A) ● (C) (D)
 14 – (A) ● (C) (D)
 15 – (A) (B) ● (D)

- 16 – (A) ● (C) (D)
 17 – (A) (B) ● (D)
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 22 – (A) ● (C) (D)
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 24 – (A) (B) (C) ●
 25 – (A) ● (C) (D)
 26 – (A) (B) ● (D)
 27 – (A) (B) (C) ●
 28 – ● (B) (C) (D)
 29 – ● (B) (C) (D)
 30 – (A) (B) (C) ●

Marker's use only.

QUESTION	1-15	16-25	27-30	Total	%
MARKS	15	10	5	30	